

RBE 500 – Homework 2

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1 Part 1: Problem Set

Problem 2-35: Verify Equation (2.67).

$$H^{-1} = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix}$$

If $H = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix}$, we can verify its inverse Equation (2.67) by proving the dot product of the equations equals an identity matrix.

$$\begin{aligned} H \cdot H^{-1} &= \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} RR^T & -RR^T d + d \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} I & 0 \\ 0 & 1 \end{bmatrix} \\ &= I \end{aligned}$$

Problem 2-36: Compute the homogenous transformation representing a translation of 3 units along the x-axis followed by a rotation of $\frac{\pi}{2}$ about the current z-axis followed by a translation of 1 unit along the fixed y-axis. Sketch the frame. What are the coordinates of the origin o_1 with respect to the original frame in each case?

While taking account of the order of matrix multiplication, current frames are post-multiplied and fixed frames are pre-multiplied. With that, we have:

$$H = T_{y,1} \cdot T_{x,3} \cdot R_{z,\frac{\pi}{2}}$$

The individual matrices are:

$$T_{x,3} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_{z,\frac{\pi}{2}} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T_{y,1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Computing $T_{x,3} \cdot R_{z,\frac{\pi}{2}}$:

$$T_{x,3} \cdot R_{z,\frac{\pi}{2}} = \begin{bmatrix} 0 & -1 & 0 & 3 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Then pre-multiplying by $T_{y,1}$:

$$H = \begin{bmatrix} 0 & -1 & 0 & 3 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The origin o_1 expressed in the base frame is the translation vector (last column):

$$o_1 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$$

Problem 2-37: Consider the diagram of Figure 2.13. Find the homogenous transformations H_1^0 H_2^0 H_2^1 representing the transformations among the three frames shown. Show that $H_2^0 = H_1^0 H_2^1$.

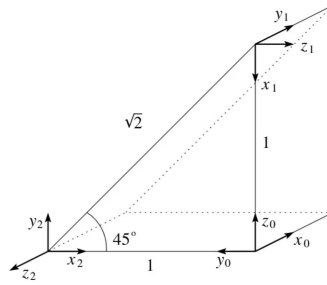


Figure 2.13: Diagram for Problem 2-37.

Frame 1 is 1 unit above Frame 0 along z_0 , with axes: $x_1 = -z_0$, $y_1 = y_0$, $z_1 = x_0$.

Frame 2 is 1 unit along y_0 from the origin, with axes: $x_2 = -y_0$, $y_2 = z_0$, $z_2 = -x_0$.

$$H_1^0 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad H_2^0 = \begin{bmatrix} 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverse of H_1^0 :

$$(H_1^0)^{-1} = \begin{bmatrix} 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Transformation H_2^1 :

$$\begin{aligned}
 H_2^1 &= (H_1^0)^{-1} \cdot H_2^0 \\
 &= \begin{bmatrix} 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & -1 & 0 & -1 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

Problem 2-38: Consider the diagram of Figure 2.14. A robot is set up 1 meter from a table. The table top is 1 meter high and 1 meter square. A frame $o_1x_1y_1z_1$ is fixed to the edge of the table as shown. A cube measuring 20 cm on a side is placed in the center of the table with frame $o_2x_2y_2z_2$ established at the center of the cube as shown. A camera is situated directly above the center of the block 2 meters above the table top with frame $o_3x_3y_3z_3$ attached as shown. Find the homogenous transformation relating the frame $o_2x_2y_2z_2$ to the camera frame $o_3x_3y_3z_3$.

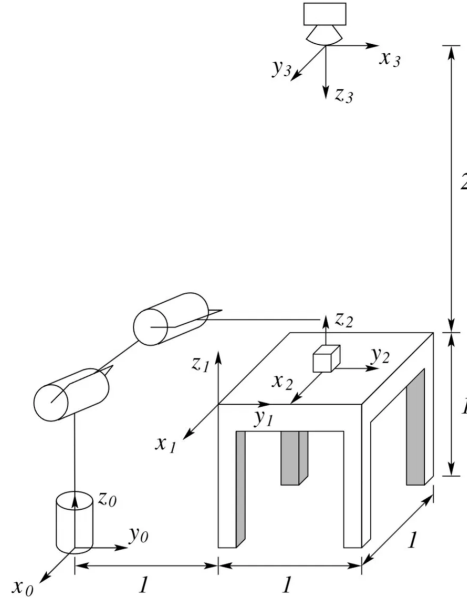


Figure 2.14: Diagram for Problem 2-38.

From Figure 2.14, we identify the following frames relative to Frame 0 (robot base). The robot is 1m from the table edge, the table is 1m high and 1m square, the cube is 20cm on a side (so its center is 0.1m above the table), and the camera is 2m above the table top.

- Frame 1: table edge, translated 1m along y_0 and 1m up along z_0 , same orientation as Frame 0
- Frame 2: cube center, offset -0.5m along x_0 , 1.5m along y_0 , and 1.1m along z_0 (1m table + 0.1m half-cube height), same orientation as Frame 0
- Frame 3: camera, directly above cube center at 3m height, same orientation as Frame 0

Homogeneous transformations relative to Frame 0:

$$H_1^0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad H_2^0 = \begin{bmatrix} 1 & 0 & 0 & -0.5 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & 1.1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad H_3^0 = \begin{bmatrix} 1 & 0 & 0 & -0.5 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Finding $H_2^3 = (H_3^0)^{-1} \cdot H_2^0$:

Since Frame 3 has no rotation relative to Frame 0 (i.e. $R_3^0 = I$), the inverse simplifies to:

$$(H_3^0)^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0.5 \\ 0 & 1 & 0 & -1.5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_2^3 = (H_3^0)^{-1} \cdot H_2^0 = \begin{bmatrix} 1 & 0 & 0 & 0.5 \\ 0 & 1 & 0 & -1.5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -0.5 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & 1.1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1.9 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The translation vector $(0, 0, 1.9)^T$ confirms that the cube center lies directly below the camera at a distance of $2 - 0.1 = 1.9$ meters, which is consistent with the geometry in Figure 2.14.

Problem 2-42: In general, multiplication of homogenous transformation matrices is not commutative. Consider the matrix product

$$H = R_{x,\alpha} \cdot T_{x,b} \cdot T_{z,d} \cdot R_{z,\theta}$$

Determine which pairs of the four matrices on the right-hand side commute. Explain why these pairs commute. Find all permutations of these four matrices that yield the same homogenous transformation matrix, H .

The four matrices are:

$$R_{x,\alpha} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_\alpha & -s_\alpha & 0 \\ 0 & s_\alpha & c_\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_{x,b} = \begin{bmatrix} 1 & 0 & 0 & b \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{z,d} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R_{z,\theta} = \begin{bmatrix} c_\theta & -s_\theta & 0 & 0 \\ s_\theta & c_\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Commuting Pairs:

- $R_{x,\alpha}$ and $T_{x,b}$ **commute**: a rotation about the x -axis and a translation along the x -axis act independently.
- $T_{x,b}$ and $T_{z,d}$ **commute**: pure translations always commute regardless of direction.
- $T_{z,d}$ and $R_{z,\theta}$ **commute**: a translation along the z -axis and a rotation about the z -axis act independently.
- All other pairs **do not commute**.

Valid Permutations:

Using the commuting pairs to swap neighbors, the following permutations all yield the same H :

1. $R_{x,\alpha} \cdot T_{x,b} \cdot T_{z,d} \cdot R_{z,\theta}$
2. $T_{x,b} \cdot R_{x,\alpha} \cdot T_{z,d} \cdot R_{z,\theta}$
3. $R_{x,\alpha} \cdot T_{x,b} \cdot R_{z,\theta} \cdot T_{z,d}$
4. $T_{x,b} \cdot R_{x,\alpha} \cdot R_{z,\theta} \cdot T_{z,d}$
5. $R_{x,\alpha} \cdot T_{z,d} \cdot T_{x,b} \cdot R_{z,\theta}$

2 Part 2: ROS 2 Assignment

2.1 Overview

Brief description of what the node does.

2.2 Package Structure

```
quat_to_euler/  
├── quat_to_euler/  
│   ├── __init__.py  
│   └── quat_to_euler_node.py  
├── package.xml  
├── setup.py  
└── setup.cfg
```

2.3 Node Implementation

```
1 # Paste your Python code here  
2 import rclpy  
3 from rclpy.node import Node  
4 ...
```

2.4 Conversion Formulas

The quaternion (w, x, y, z) is converted to Euler angles (ϕ, θ, ψ) using:

$$\phi = \arctan 2(2(wx + yz), 1 - 2(x^2 + y^2)) \quad (1)$$

$$\theta = \arcsin(2(wy - zx)) \quad (2)$$

$$\psi = \arctan 2(2(wz + xy), 1 - 2(y^2 + z^2)) \quad (3)$$

2.5 How to Run

```
ros2 run quat_to_euler quat_to_euler_node
```

Publish a test quaternion from the terminal:

```
ros2 topic pub /quaternion std_msgs/msg/Float32MultiArray \  
"data: [1.0, 0.0, 0.0, 0.0]"
```

2.6 Results